

Pathway to carrageenan-induced inflammation in the hind limb of the rat¹

RALPH VINEGAR, JAMES F. TRUAX, JEFFREY L. SELPH, PHILIP R. JOHNSTON, ALEA L. VENABLE, AND KERRY K. MCKENZIE

Department of Pharmacology, Wellcome Research Laboratories, Research Triangle Park, North Carolina 27709

A sequential 12-step pathway scheme representing the development of carrageenan (C) inflammation in the rat was presented at the first symposium of this series (17). The stimulus for these studies was the ease with which quantitative estimation of edema could be made from changes in hind-limb volume. Temporal studies revealed a biphasic edematous response in the first 3 h. Steroidal and nonsteroidal antiinflammatory drugs inhibited the development of the second phase, but did not affect the first phase. When C pleurisy was introduced (4) our efforts were diverted to studies of this model of inflammation because quantitative temporal measurements of the number of mobilized phagocytic cells as well as quantitative estimation of exudate volume could be made in the rat (16, 19). A sequential 37-step pathway scheme was proposed for the C pleurisy model (20). It consisted of a dual phagocytic inflammatory response (PIR). The site of the primary reaction was observed histologically to be the diaphragmatic pleura. The secondary response was intrapleural exudation. Although exudation followed the inflammation in the pleura temporally, it was not produced by a simple overflow of cells and fluid from the primary site (20). The potential difficulties of delineating a sequential pathway scheme for a dual inflammatory response led us to return to studies of the subplantar, interdermal, response to C in the rat.

HISTOLOGICAL STUDIES

In previous experiments, quantitative hind-limb volume displacement measurements revealed the presence of appreciable edema 20 min after the

ABSTRACT

A sequential 43-step pathway scheme for the inflammatory response of the rat to interdermal injection of carrageenan (C) was devised. It consisted of a nonphagocytic inflammatory response (NPIR) followed by a phagocytic inflammatory response (PIR) in the dermis and an epidermal NPIR. The dermal NPIR comprised edema, hyperemia, and hyperalgesia followed by hypoalgesia. Antiserotonin agents inhibited the hypoalgesia and part of the edema. These findings and histological observations suggested that dermal mast cells were injured by C. The hyperalgesia and part of the edema were sensitive to arachidonate cyclooxygenase inhibitors (AACOIs). It is speculated that injured mast cells metabolize arachidonic acid and reactive intermediates, not prostaglandins, mediate the NPIR hyperalgesia and part of the edema. The dermal PIR consisted of mobilization of neutrophils, edema, hyperalgesia, mobilization of monocytes, and proliferation of fibroblasts and vascular tissue. Selective drug actions revealed that the edema, hyperalgesia, and monocyte mobilization of the PIR depended on the mobilization of neutrophils. After the mobilization of neutrophils, AACOIs reduced edema formation and hyperalgesia. Arachidonic acid metabolism by neutrophils is speculated to produce the mediators of phagocytic inflammatory (PI) edema and hyperalgesia. Monocyte function was associated with cessation of PI edema formation and phagocytosis of neutrophils and cellular debris. Interleukin 1 is speculated to mediate the adherence of neutrophils to injured dermal endothelium. The epidermal NPIR consisted of edema, hyperplasia, and hyperkeratosis. These parameters were not studied mechanistically. There was no evidence for histamine, bradykinin, platelets, clotting factors, or complement mediating any events in the pathway—Vinegar, R.; Truax, J. F.; Selph, J. L.; Johnston, P. R.; Venable, A. L.; McKenzie, K. K. Pathway to carrageenan-induced inflammation in the hind limb of the rat. *Federation Proc.* 46: 118–126; 1987.

subplantar injection of 500 μ g of C in the rat (14). Histological examination of dermal and epidermal tissue removed at this time did not detect any cellular changes compared to uninjected tissue. Histological observations made on tissue removed 180 min after the injection of C disclosed an intense PIR in the dermis consisting of large numbers of neutrophils and tissue edema. From the observations just cited it was evident that a nonphagocytic inflammatory response (NPIR) preceded development of the PIR. To separate the events associated with the

NPIR from those of the PIR, a temporal study was undertaken in which quantitative hind-limb edema measurements were combined with histological observations (Table 1). Neu-

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TABLE 1. Early development of C inflammation in the rat hind limb^a

Drug	Dose, mg/kg orally	Time, min	Hind-limb edema volume, ml $\pm$ SEM	Relative number of N present in subplantar tissue, mean			
				Lumen	Intramural	Extramural	
						Perivascular	Interstitial
None	—	0	—	0	0	0	0
NaCMC	50	20	0.19 $\pm$ 0.026	0	0	0	0
NaCMC	50	60	0.29 $\pm$ 0.025	0	0	0	0
NaCMC	50	180	0.83 $\pm$ 0.043	2.1	2.1	2.2	0.9

^a There were seven rats in each group. The first group was untreated and received no subplantar injection. Each rat in the remaining groups received a single subplantar injection of 500 μ g of C in 0.05 ml of pyrogen-free water. Rats were killed at various times after injection of C. A strip of tissue approximately 1.0 cm long and 0.5 cm wide surrounding the injection site was fixed in Bouin solution, embedded in paraffin, and stained with hematoxylin and eosin. The number of neutrophils (N) present at each location was estimated by light microscopic examination, using an arbitrary scoring system of 0 (rarely observed) to 4 (moderate infiltration). The observer was unaware of the treatment of the animals. CMC represents carboxymethylcellulose.

trophils were not observed in tissue sections taken 20 and 60 min after the injection of C. At 180 min edema and large numbers of neutrophils were observed histologically in and around the small blood vessels in the dermis. The time of onset of neutrophil mobilization, which represents the beginning of the PIR, is assumed, therefore, to be 60 min. All events occurring in the dermal and epidermal tissue before 60 min are categorized as components of the NPIR.

Appreciable nonphagocytic edema was measured 20 and 60 min after the subplantar injection of C (Table 1), when neutrophils were not present in histological sections. The time course of development of this hind-limb edema was studied as a function of the dose of C (Fig. 1). The edematous response consisted of three components that developed in the 0- to 50-min interval after the injection of C. The first component developed in the 0- to 2-min interval and was small in absolute value, 0.04 ml. This component was produced by each dose of C or by the solvent. It is assumed to result from direct physical damage to small blood vessels at the injection site by the inoculum. Because this component is produced by injection of the solvent, it is not considered part of the response to C and is not included in the subsequent pathway scheme. The second component of the nonphagocytic edema developed in the 2- to 10-min interval (Fig. 1). It was not produced by the solvent. Its magnitude was small, about 0.15 ml, and independent of the dose of C in the range tested. This component was reduced in animals pretreated with the arachidonate

cyclooxygenase inhibitors (AACOIs) aspirin or ibuprofen and the antiserotonin agents methysergide or cyproheptadine, as shown in Table 2. High dose levels of these inhibitors did not produce more than 60% inhibition, which suggests that about half of this edema is produced by serotonin and the other half by an arachidonic acid metabolite. This analysis is reinforced by the ability of a combination of aspirin and methysergide to block the

development of this component of the edema. The third component of the nonphagocytic edema developed in the 10- to 50-min interval (Fig. 1). Its magnitude was a function of the dose of C. Development of this edema was not inhibited by pretreatment with steroidal or nonsteroidal antiinflammatory drugs, antihistaminics, or antiserotonin agents. The etiology of this component is unknown. For the purpose of building a pathway scheme, we assume that it results from osmotic forces. The magnitude of these forces is a function of the concentration of C. These physicochemical forces are not affected by drug treatment.

An agent that can be used to resolve the edemas of a NPIR or PIR was found to be cycloheximide. When injected in the subplantar tissue in combination with C, cycloheximide did not reduce the magnitude of the nonphagocytic edema, but it blocked the phagocytic edema (Fig. 2). This selective action made it possible to determine the magnitude of edema produced by the PIR (termed the cycloheximide-sensitive edema in Table 2). Steroidal and nonsteroidal antiinflammatory drugs inhibited the develop-

Figure 1. Temporal development of nonphagocytic hind-limb edema as a function of dose of C. The volume of injection, 0.10 ml, has not been subtracted from the edema volumes. There were six rats in each group. Different groups of rats were used for each determination.

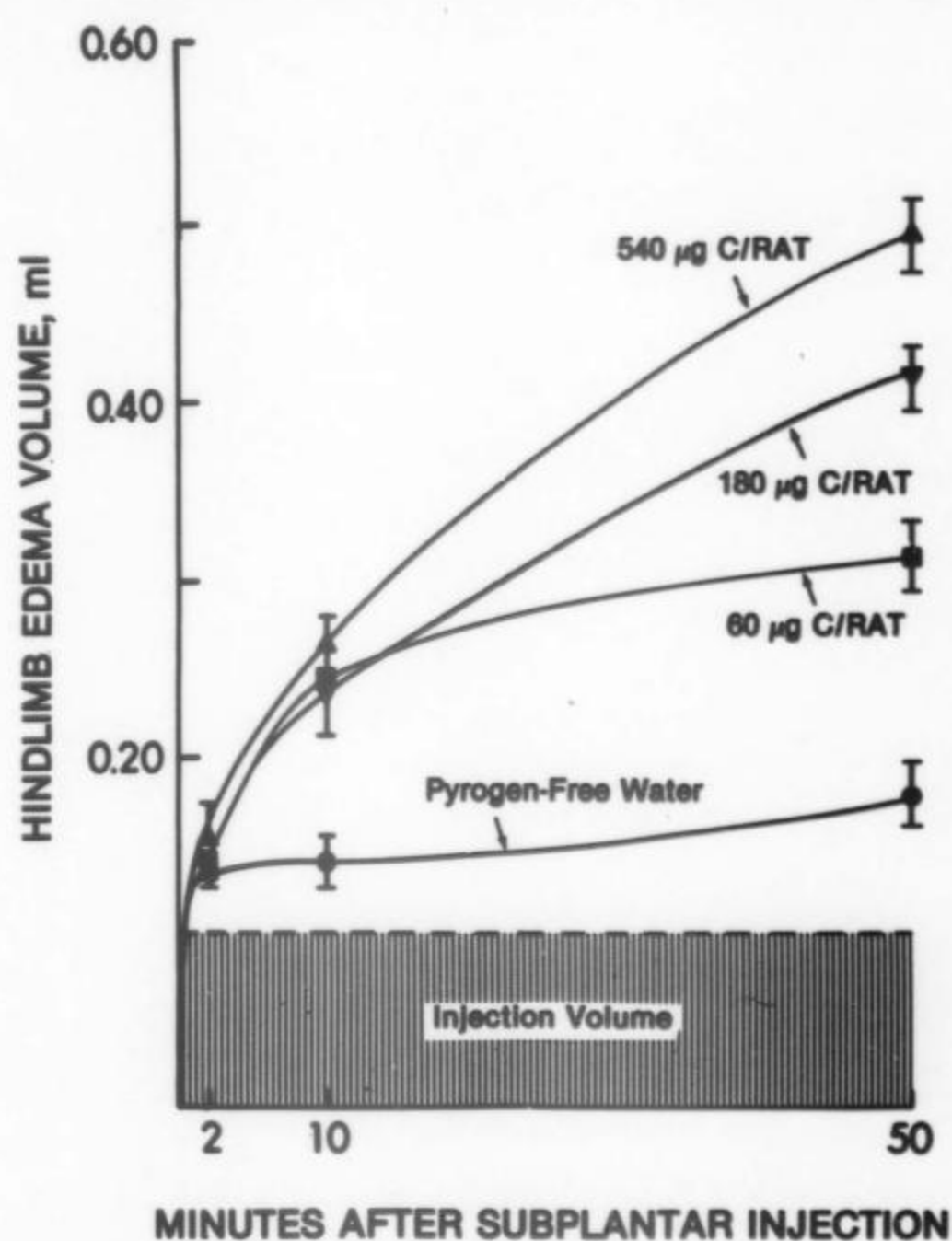


TABLE 2. Drug effects on early development of C hind-limb edema in the rat^a

Drug	Inhibition of C hind-limb edema, ED ₅₀ , mg/kg orally		
	10-2 min ^b	50-10 min	Cycloheximide-sensitive edema at 180 min
Acetylsalicylic acid	46	I at 100	30
Ibuprofen	10	I at 100	9.9
Hydrocortisone	I at 100	I at 100	10
Prednisolone	I at 10	I at 10	1.0
Betamethasone	I at 0.10	I at 0.10	0.05
Chlorpheniramine	>10	I at 10	I at 10
Cyproheptadine	8.0	I at 10	I at 10
Methysergide, i.p.	0.69	I at 1.0	I at 1.0

^a Each rat received a single subplantar injection of 500 μ g of carrageenan in 0.10 ml of pyrogen-free water. Each experiment contained a positive and a negative control. Drugs were administered 30-60 min before measurement of each phase. ED₅₀ values for each drug were estimated against each phase of C hind-limb edema by regression of at least three dose levels on percent inhibition of the hind-limb edema volume. There were at least five rats at each dose level. I represents inactive. ^b Minutes after injection.

ment of the phagocytic edema, whereas antihistaminics and antisero-tonin agents had no effect.

ADHESION OF NEUTROPHILS TO ENDOTHELIUM

In recent publications irritants such as endotoxin (23, 24), interferon (6), and interleukin 1 (IL 1) (2, 3) were reported to increase the adhesivity of monolayers of endothelial cells in culture to phagocytic cells or B and T lymphocytes. Actinomycin D and cycloheximide reduce the adhesiveness of human neutrophils produced by IL 1 on human umbilical vein endothelial cell monolayers (2). As a first approach in extending these in vitro observations to an experimental in vivo system we used the C pleurisy model in the rat (16). In this model drugs can be injected directly into the pleural cavity before or along with the irritant, and the subsequent number of neutrophils mobilized can be determined quantitatively. A decrease in the number of neutrophils mobilized by an agent administered locally at a dose that is inactive when injected by systemic routes is assumed to be a measure of reduced endothelial adhesivity. Histological examination of the site of irritation can confirm the absence of neutrophils in the irritated tissue. When injected in the subplantar tissue in combination with C, cycloheximide and actinomycin D reduced the number of neutrophils mobilized. The dose of cycloheximide required locally to reduce the number of neutrophils mobilized by 50% was equal, statistically, to the dose required by i.p. injection (Table

3). This equality indicates that the activity of cycloheximide after subplantar injection resulted from a systemic action. In contrast, actinomycin D was 250-500 times more potent when injected locally than when injected i.p.

(Table 3). This wide separation in potency favoring local administration strongly suggested that actinomycin D acts at the injection site to reduce the number of neutrophils mobilized, presumably by reducing the adhesivity of the irritated endothelium for these phagocytes (2). Microscopic examination of sections of the parietal pleura revealed few neutrophils in the irritated tissue of animals injected with actinomycin D. This observation contrasted sharply with the large number of neutrophils in the vessels and irritated tissue of control animals injected with C. Actinomycin D is considered an antiinflammatory agent by this action. AACOIs have not demonstrated the ability to reduced the number of neutrophils mobilized when injected locally in this model (Table 3). However, dose levels of actinomycin D that produced specific local activity also decreased the number of free resident cells in the pleural cavity from 7.1 million to 3.6 million on average. This

Figure 2. Local action of cycloheximide (CHI) on C hind-limb edema in the rat. Subplantar injections of C, cycloheximide, or C and cycloheximide were given. Hind-limb edema volumes were then measured as a function of time after injection of these agents. The volume of injection, 0.10 ml, has not been subtracted from the edema volumes. There were six rats in each group.

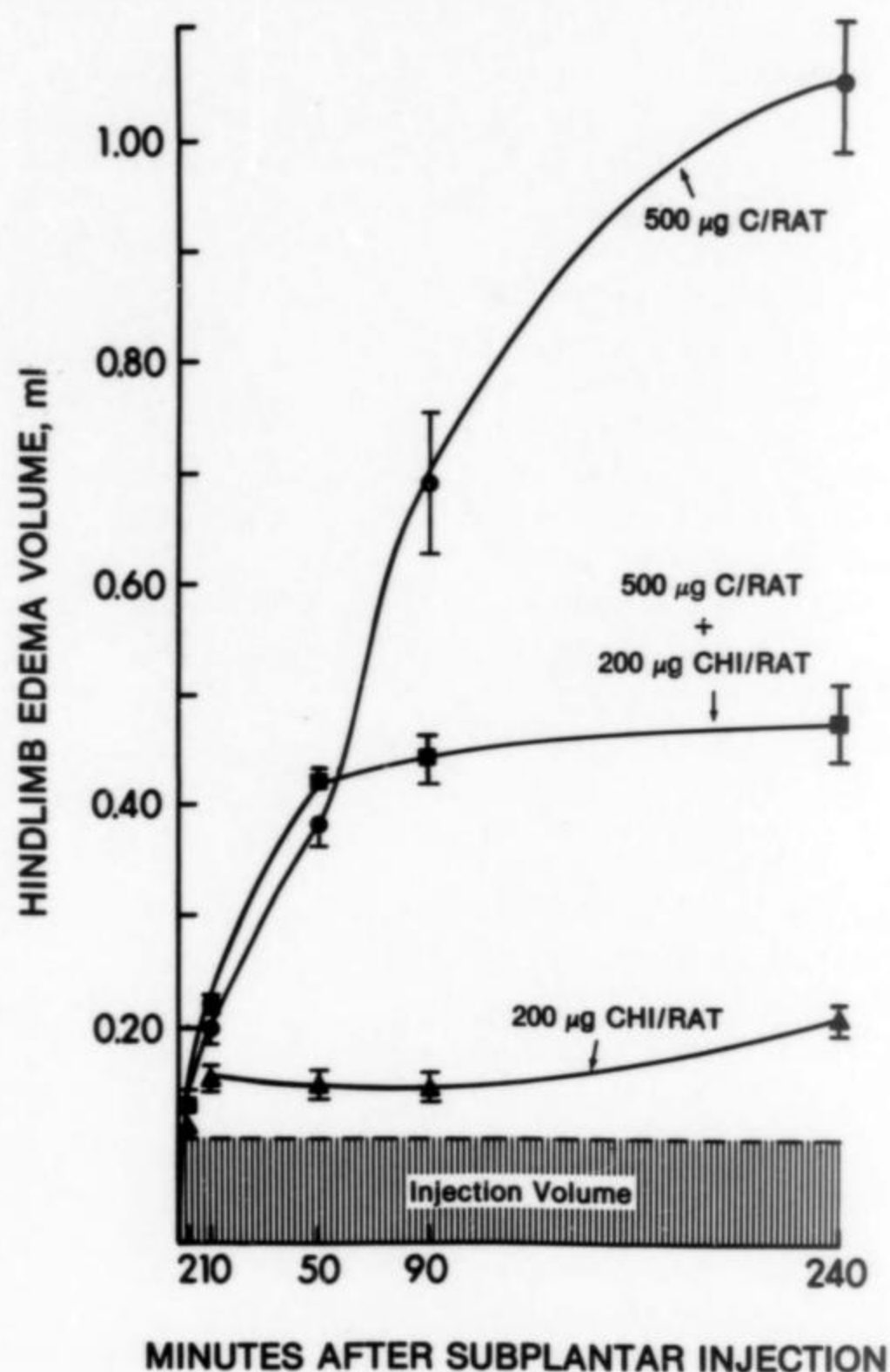


TABLE 3. Drug actions in the C pleurisy assay in the rat^a

Drug	C pleurisy assay—3 h, ED ₅₀ , mg/kg			
	i.p.		Local (intrapleural)	
	Exudate volume	Neutrophil mobilization	Exudate volume	Neutrophil mobilization
Aspirin	25	23	0.27	I at 1.3
Indomethacin	1.3	12	0.017	I at 0.07
Prednisolone	3.7	3.5	0.078	0.13
Betamethasone	0.18	0.41	0.011	0.006
Cycloheximide	0.21	0.17	0.093	0.16
Actinomycin D	0.50	0.50	0.0020	0.0010

^a Each rat received an intrapleural injection of 150 μ g of C in 0.10 ml of pyrogen-free water. Drugs were administered i.p. 30 min before the injection of C. Aspirin and indomethacin were given intrapleurally 1 h after the injection of C and all other drugs were given with the C to produce peak effects. Pleural exudates were harvested 3 h after C injection. ED₅₀ values were estimated from regression analysis of at least three dose levels of each drug. There were five rats in each group. I represents inactive.

action indicates a form of cytotoxicity and occurred when actinomycin D was injected alone or in combination with C. Actinomycin D may also be toxic to endothelium in vivo, although cytotoxicity was not observed on endothelial cell monolayers in vitro (2). The relationship of the in vivo cytotoxicity of actinomycin D to its reduction of neutrophil mobilization is not known. We reported previously (20) that washout of 87% of the resident cells of the pleural cavity of the rat did not affect the magnitude of exudate volume or cell mobilization elicited by a subsequent injection of C. However, a washout procedure cannot be equated to a cytotoxic action because the latter may result in release of destructive cellular enzymes and cytokines into the pleural space.

In our first pathway scheme representing C-induced inflammation, 2 of 12 steps concerned a chemotactic factor that past dogma theorized would be found in the circulating blood. We could not find an agent with in vivo chemotactic activity in the plasma or edema fluid of animals injected with C (17, 20). In addition, when we injected cobra venom factor and waited for its irritancy to subside there was no reduction in the number of neutrophils mobilized by C although levels of complement were reduced by more than 90% (17). These past findings and the results just presented concerning actinomycin D led us to accept the concept that irritant-induced endothelial cell adhesivity is responsible for the localization of phagocytes at the site of an irritation. The presence of IL 1 in many types of cells including vascular endothelium (7), its expres-

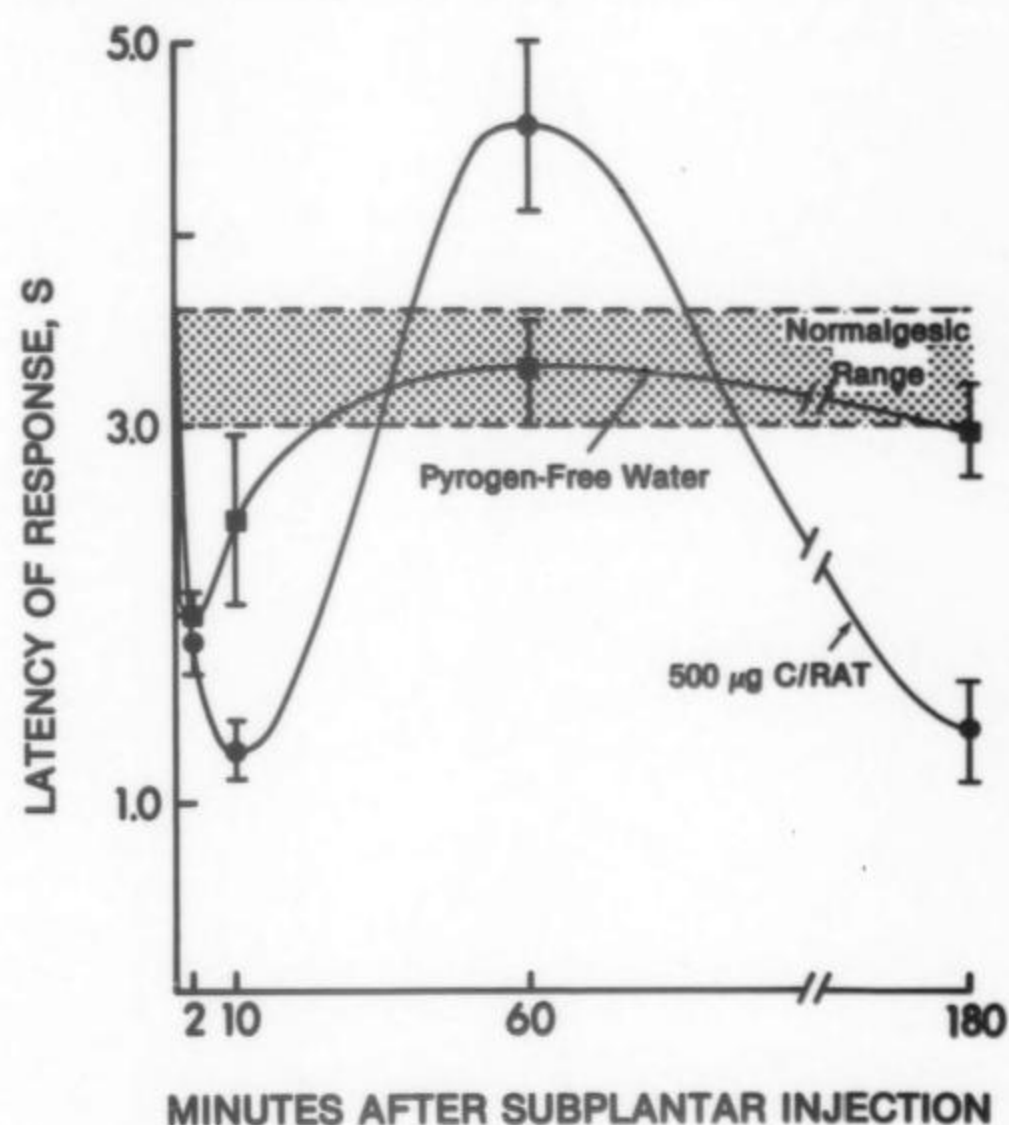
sion by a wide variety of irritants (1), and its ability to increase the adhesiveness of endothelial cells in culture for human and animal phagocytes have made this cytokine a prime candidate for mediating the localization of phagocytic cells at sites of irritation. This proposed mediation by IL 1 will be used in the subsequent pathway scheme. It is understood that more experimental evidence is needed to firmly establish this mechanism. We previously illustrated the weakness of assuming that the presence of a potential mediator at a site is sufficient evi-

dence for its participation in a specific event at that site (15).

ALGESIC CHANGES PRODUCED BY C

An important clinical aspect of many inflammatory diseases is pain and soreness. In animal models these parameters are correlated with increased sensitivity of inflamed tissue to painful stimuli—hyperalgesia (10, 18). In some model systems AACOIs are able to reduce the hyperalgesia that develops. These reductions are dose-related and presumed to be a measure of the clinical analgesic activity of these drugs (18). At relatively high dose levels, some prostaglandins (PGs) produced short-lived hyperalgesia in the rat hind limb. Because these high dose levels are not likely to be attained in vivo, we proposed that the active hyperalgesic principle is a reactive PG intermediate (i.e., endoperoxide). The complexity of the nonphagocytic edema resulting from the injection of C led us to investigate the changes in algesia that occur as each of the three components of this response develops (Fig. 3). Moderate hyperalgesia was measured 2 min after the injection of C or its solvent, pyrogen-free water.

Figure 3. Time course of C hyperalgesia in the rat. The average latency (seconds) of a group of rats in response to a 2.0-kg force applied to each hind limb was measured at various times after subplantar injection of 500 μ g of C or the solvent, pyrogen-free water. The normal algesic range (latency $\pm$ SEM of a group of untreated naive rats) is represented by the stippled area. A different group of rats was used for each determination ($n = 6$).



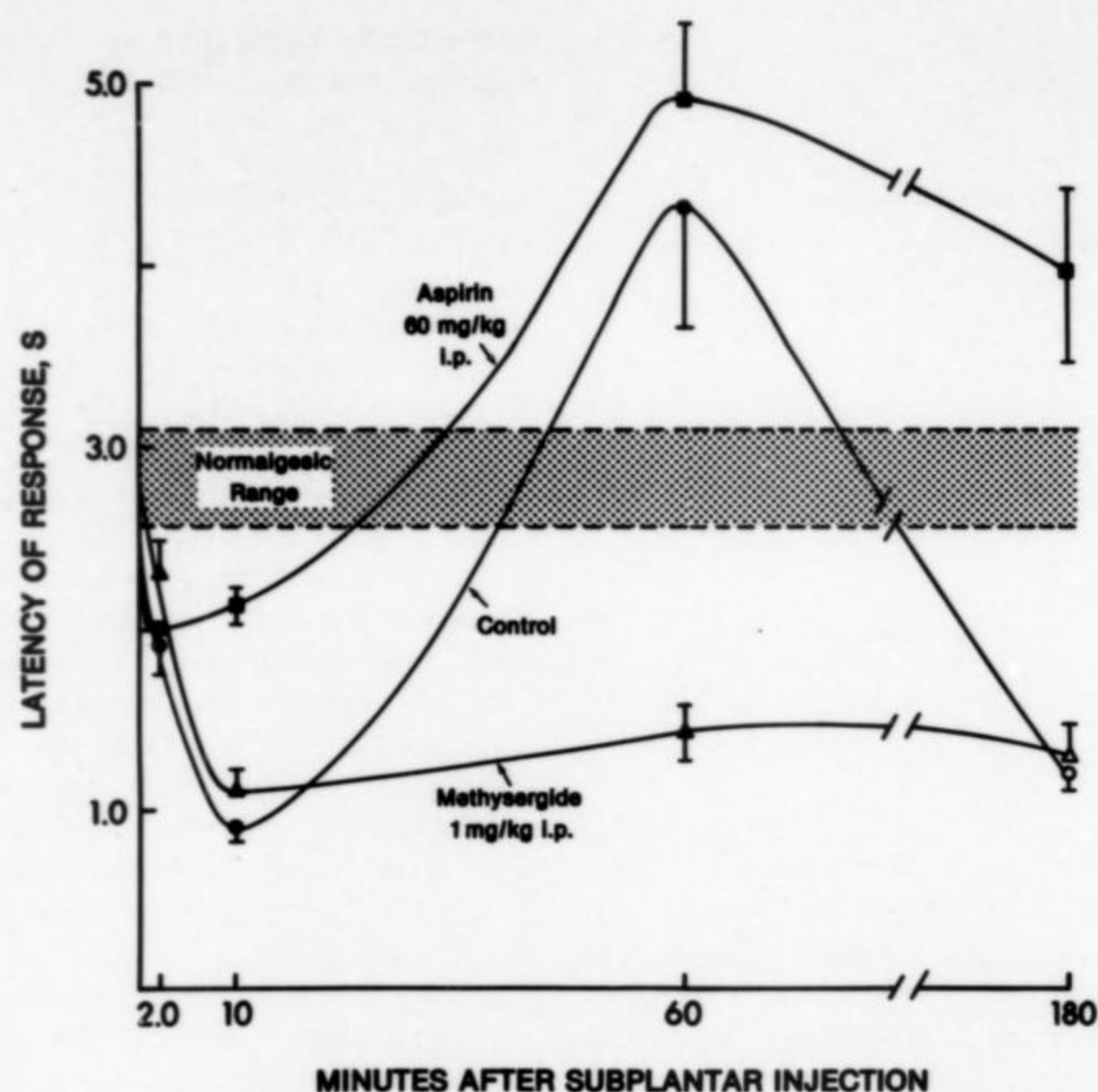


Figure 4. Drug effects on the early development of C hyperalgesia in the rat. The average latency (seconds) of a group of drug- or solvent-treated rats in response to a 2.0-kg force applied to each hind limb was measured at various times after the subplantar injection of 500 µg of C. Drugs were administered 30 min before C injection. The normalalgesic range is represented by the stippled area. There were five rats in each group. Each group was used for one determination.

This hyperalgesia was short-lived. It decayed to the normalalgesic level in less than 60 min. It is not considered to be part of the algesic response to C. At 10 min, the end of the development of the second component of the non-phagocytic edema, a peak of hyperalgesia was measured in animals injected with C. This hyperalgesia is short-lived. It is overcome by an intense hypoalgesic component that began to develop at 10 min and peaked at 60 min. The temporal development of hypoalgesia parallels the time course of the third component of the non-phagocytic edema (Fig. 1). Both develop in the 10- to 50-min interval. Finally, 180 min after the injection of C another phase of hyperalgesia was measured (Fig. 3). This late phase of hyperalgesia was previously shown to depend on mobilization of neutrophils for its development (18) and therefore is part of the PIR.

The mediators responsible for the algesic changes that occurred during the NPIR to C was investigated by specific pharmacological intervention (Fig. 4). Aspirin reduced the hyperalgesia present at 10 min without af-

fecting the subsequent development of hypoalgesia. Methysergide did not affect the hyperalgesia produced at 10 min, but blocked the development of hypoalgesia in the 10- to 60-min interval. This action of methysergide is consonant with the reported hypoalgesia resulting from the subplantar injection of serotonin in the hind limb of the rat (13, 22). In those studies the

dose of serotonin was 10 µg per hind limb. This dose of serotonin produces a large edema in the hind limb, approximately 1.2 ml, that antiserotonin agents can inhibit (5, 11). However, the edema produced by C in the 10- to 50-min interval was not affected by antiserotonin agents (Table 2). These apparently conflicting findings led us to consider the possibility that low levels of serotonin can produce significant changes in the level of algesia without eliciting a concomitant edema. This possibility was investigated by injecting low dose levels of serotonin and measuring the edema formed and the algesic changes (Fig. 5). A dose of 0.020 µg of serotonin did not produce a measurable edema. A very small edema, 0.08 ml, was measured 10 and 60 min after the injection of 0.10 µg of serotonin. These low dose levels of serotonin produced hyperalgesia 10 min after injection that was followed by a long-lived hypoalgesic component (Fig. 5). Thus, low dose levels of serotonin can produce algesic changes without significant edema formation. The magnitude of the algesic changes produced by low dose levels of serotonin was dose-related. At 10 and 60 min the changes produced by 0.10 µg of serotonin were in the range of those measured after the injection of 500 µg of C (Fig. 5). These similarities suggested that injection of C may result in injury to dermal mast cells, which then express their stores of serotonin. This possibility was investigated by a comparison of the sensitivities of the hyperalgesia and hypoalgesia elicited by C and serotonin to a variety of pharmacological agents (Table 4).

TABLE 4. Effect of various drugs on C- and serotonin-induced algesia in the rat hind limb^a

Drug	ED ₅₀ , mg/kg, orally			
	Hyperalgesia (10 min)		Hypoalgesia (60 min)	
	Carrageenan	Serotonin	Carrageenan	Serotonin
Aspirin	11	22	I at 60	I at 60
Ibuprofen	16	23	I at 60	I at 60
Prednisolone	I at 10	I at 10	I at 10	I at 10
Betamethasone	I at 1.0	>0.20	I at 1.0	I at 1.0
Chlorpheniramine	I at 10	I at 10	I at 10	I at 10
Cyproheptadine	I at 10	>10	4.2	5.8
Methysergide	I at 1.0	I at 1.0	0.56	0.50

^a Drugs were administered by gavage in 10 ml/kg of body weight. Thirty minutes later each rat received a single subplantar injection of 500 µg of C or 0.10 µg of serotonin in 0.10 ml of pyrogen-free water. Latencies of response to the application of a 2.0-kg force to the injected hind limb were measured 10 min (hyperalgesia) and 60 min (hypoalgesia) after injection of each irritant. ED₅₀ values were estimated by regression analysis by using at least three dose levels of each drug. Each experiment included a positive and a negative control. There were five rats in each group. I represents inactive.

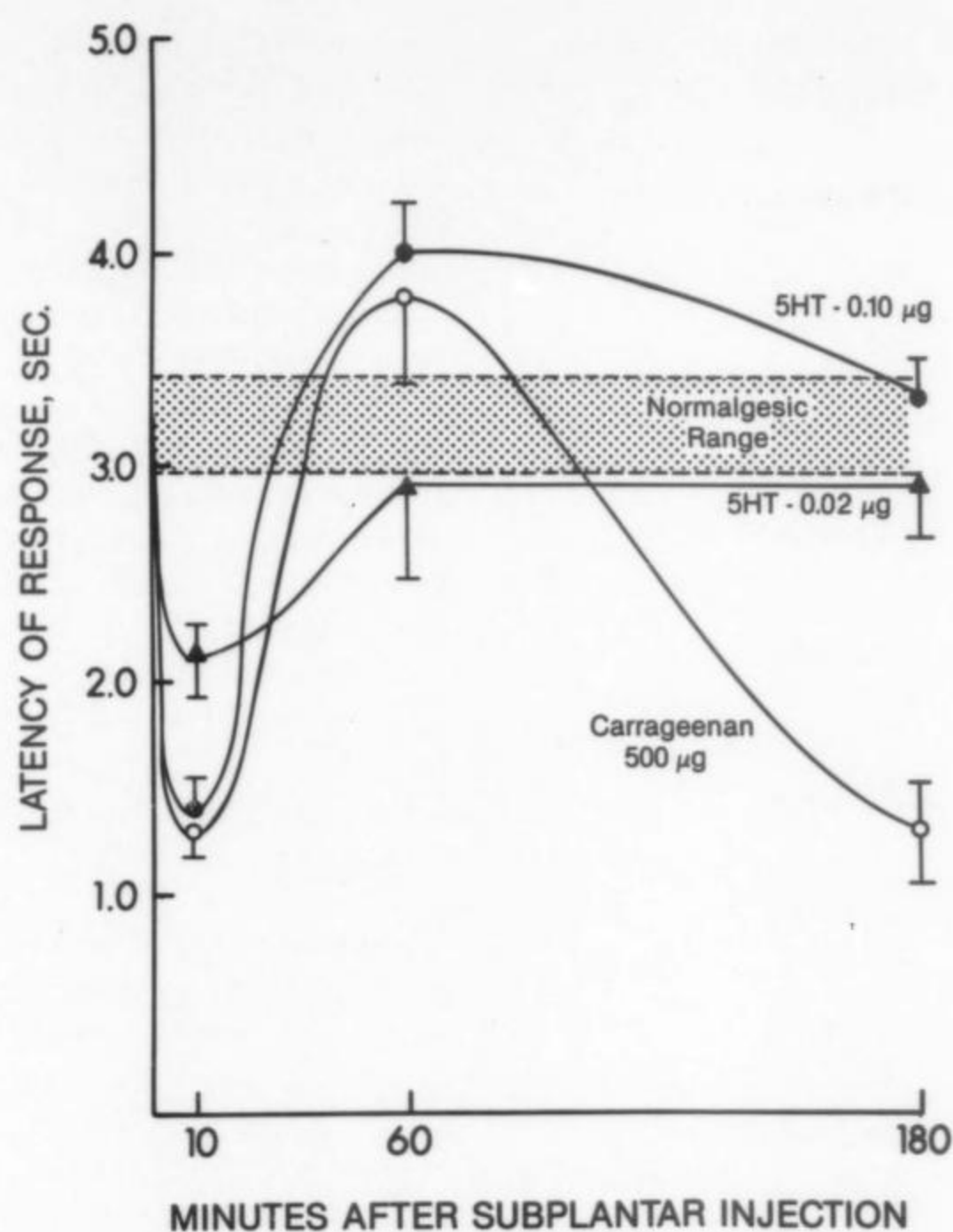


Figure 5. Comparison of time courses of serotonin- and C-induced hyperalgesia and hypoalgesia in the rat. Rats, in groups of five, each received a single subplantar injection of either serotonin or C in a volume of 0.10 ml pyrogen-free water. The average latency (seconds) of response of the rats on application of a 2.0-kg force to the irritant- or solvent-injected hind limb is measured as a function of time after irritant injection. The normalalgic range is represented by the stippled area. A different group of rats was used for each determination.

Their sensitivities to drugs were similar. The hyperalgesia they produced 10 min after injection was selectively sensitive to AACOIs. Increases in latency produced by these analgesics were dose-dependent. Equivalent ED_{50} values were obtained against the hyperalgesia of C and serotonin. The hypoalgesia produced by C and serotonin at 60 min was sensitive to antiserotonin agents (Table 4). These findings indicate that the early changes in algesia elicited by C may result from the release of tissue stores of serotonin. Although histamine is released along with serotonin, only the serotonin elicits edema formation in the rat (20).

PATHWAY SCHEME FOR C HIND-LIMB INFLAMMATION

A sequential 43-step pathway scheme has been devised to represent the events and actions responsible for the onset, development, and decay of the

Figure 6. Pathway scheme representing the development of the dermal NPIR to C.

- Step 1.** Subplantar injection of 500 ug of C, t_0 .
- Step 2.** Absorption of smaller particles of C into dermal mast cells, t_0 - t_{10} min.
- Step 3.** Cytoplasmic injury to dermal mast cells produced by C, t_0 - t_{10} min.
- Step 4.** AA metabolism initiated by injured mast cells, t_2 - t_{10} min.
- Step 5.** Hyperalgesia and 2nd component of NPI edema produced by reactive PG intermediate, t_2 - t_{10} min.
- Step 6.** Degranulation of injured dermal mast cells, t_2 - t_{10} min.
- Step 7.** Hyperemia results from dilatation of small dermal blood vessels by a direct action of serotonin, t_2 min- t_2 days.
- Step 8.** Hypoalgesia produced by a direct action of serotonin on nerve endings in dermis, t_{10} - t_{60} min.
- Step 9.** C-dependent osmotic gradient established between injection site and surrounding dermal tissue, t_{10} - t_{60} min.
- Step 10.** Gradual development of 3rd component of NPI edema in dermis due to osmotic forces exerted by gradient of C, t_{10} - t_{60} min.

inflammatory response of the rat to subplantar injection of C (Figs. 6-9). The principal sources of information were previous pathway schemes (17, 20), in vivo data presented in the early sections of this paper, histological observations by K. Militzer (8, 9), and cytological findings described by Weissmann and colleagues (21, 25). Histological findings presented in Table 1 reveal that an NPIR preceded a PIR in the dermal tissue. The NPIR consisted of dermal edema, hyperemia (8, 9), and a brief period of hyperalgesia followed by a trough of hypoalgesia (Fig. 4). A sequential 10-step pathway scheme outlines the events and actions responsible for the NPIR (Fig. 6). Initiation of the NPIR is postulated to result from rapid absorption of small particles of C by dermal mast cells (step 2) and the cytotoxic action of these particles (25) (step 3). It is then postulated that cytoplasmic enzymes released by injured mast cells initiate arachidonate metabolism (step 4). Mast cells of the rat are capable of PG synthesis (12). In the pathway scheme it is proposed that the injured mast cells synthesize the reactive PG intermediate held responsible for the AA-COI-sensitive edema (Table 2) and hyperalgesia (Table 4) produced 2-10

- Step 11. Damage to cytoplasmic organelles of endothelial cells lining dermal blood vessels by absorbed C, $t_{90}-t_{240}$ min.
- Step 12. IL-1 expressed from cytoplasm of injured endothelial cells, $t_{90}-t_{240}$ min.
- Step 13. IL-1 mediated alteration in configuration of endothelial cell surface, $t_{90}-t_{240}$ min.
- Step 14. Neutrophils adhere to altered endothelial cell surface, $t_{90}-t_{240}$ min.
- Step 15. Diapedesis of adhered neutrophils into dermal tissue, $t_{90}-t_{240}$ min.
- Step 16. Neutrophils attracted to site of tissue damage by chemotactic cytoplasmic enzyme released from injured dermal cells, $t_{90}-t_{240}$ min.
- Step 17. Neutrophils phagocytize remaining C in dermis, $t_{90}-t_{240}$ min.
- Step 18. Injury to phagocytic vacuoles of neutrophils by ingested C $t_{90}-t_{240}$ min.
- Step 19. Fusion of lysosomes with injured phagocytic vacuoles, $t_{90}-t_{240}$ min.
- Step 20. Degranulation of lysosomes within phagocytic vacuoles, $t_{90}-t_{240}$ min.
- Step 21. Lysosomal enzyme injury to membrane of phagocytic vacuoles, $t_{90}-t_{240}$ min.
- Step 22. Intracellular release of lysosomal enzymes by neutrophils, $t_{90}-t_{240}$ min.
- Step 23. Enzymatic injury to plasma membrane of neutrophils, $t_{90}-t_{240}$ min.
- Step 24. Release of lysosomal enzymes of neutrophils into dermal tissue, $t_{90}-t_{240}$ min.
- Step 25. Neutrophil phospholipase A_2 liberates arachidonic acid from dermal cells, $t_{90}-t_{240}$ min.
- Step 26. Synthesis and release of chemically reactive PG intermediate by neutrophils in dermis, $t_{90}-t_{240}$ min.
- Step 27. Reactive PG intermediate increases permeability of endothelial cells and increases sensitivity of local nerve endings, $t_{90}-t_{240}$ min.
- Step 28. Development of PI edema and hyperalgesia, $t_{90}-t_{240}$ min.

Figure 7. Pathway scheme representing the early development of the dermal PIR to C.

min after the injection of C (step 5). Degranulation of mast cells has been observed histologically in the dermis of rats injected with C (8) (step 6). Serotonin is released in this process. Low dose levels of this biogenic amine are able to produce capillary hyperemia (step 7) and hypoalgesia (Fig. 5; step 8) when injected in the hind limb of the rat. The released serotonin is held responsible for the trough of hypoal-

gesia that peaks at 60 min (Fig. 3 and Table 4) because there is a dose-related inhibition of its development by pretreatment with antiserotonin agents (Table 4). The etiology of the third component of NPI edema that develops in the 10- to 50-min interval is unknown. Its dependence on the concentration of C (Fig. 1; step 9) and its insensitivity to antiinflammatory drugs, antiserotonin agents, and an-

tihistaminics (Table 4) led to speculation that it results from osmotic forces (step 10). These physicochemical forces would not be sensitive to drugs. There is no direct evidence supporting this speculation.

Development of the PIR follows the NPIR temporally in the proposed pathway scheme. However, although this sequence was observed in the histological study (Table 1), these two reactions are presented as independent responses by the host. This independence was observed in the response of the rat to intrapleural injection of C (20). In this model there is no NPIR associated with the exudative PIR. The early development of the PIR spans the 1- to 4-h interval after the injection of C (Fig. 7; steps 11-28). Initiation of the PIR is proposed to result from the cytotoxicity of C absorbed by the dermal endothelium (step 11) and expression of IL 1 by these injured cells (7) (step 12). The released IL 1 is speculated to alter the surface configuration of the injured dermal endothelium (step 13), which becomes adhesive for circulating neutrophils (step 14). Inhibition of C-induced neutrophil mobilization by actinomycin D (Table 3) and histological observations that this drug prevented neutrophils from entering the irritated tissue after C injection are evidence favoring this speculation (see section on adhesion). The diapedesis of neutrophils was observed in our histological studies and in those of others (8, 9) (step 15). The agent responsible for attracting the neutrophils to the injured dermal cells (step 16) is unknown. However, we previously reported that several enzymes (i.e., thrombin or collagenase) are chemotactic for neutrophils in vivo (20) and we therefore proposed that the chemoattractant is a cytoplasmic enzyme released from the injured dermal cells. Neutrophils phagocytize C, and in the process they are irreversibly damaged (24) (steps 16-24). One of the enzymes released by the injured neutrophils is speculated to be phospholipase A_2 (step 25), which released arachidonic acid from tissue phospholipids. A reactive PG intermediate is synthesized from the liberated arachidonate by the neutrophils and released (20) (step 26). This reactive PG intermediate increases the permeability of the dermal endothelium and sensitizes the nerve endings in the dermis (step 27), which

results in the PI edema and hyperalgesia (step 28). Selective inhibition of these neutrophil-dependent actions by AACOIs (Tables 3 and 4) led to these steps.

The monocyte phase of the pathway scheme initiates the healing process. This concept was garnered from our studies of C-induced pleurisy in the rat (20). In these studies the mobilization of monocytes was associated with phagocytosis of neutrophils and cessation of edema formation. In the hind-limb studies our observations are consonant with those of the pleurisy model. Monocytes are mobilized to the inflamed area by an enzyme released from the injured neutrophils, which alters the outer surface of the dermal endothelium so that it is adhesive for monocytes (step 29). The rationale for this speculation is discussed in a prior pathway paper (20). Monocytes begin to appear in the dermis at 7 h and peak in number at about 16 h (8) (steps 30, 31). It is speculated that they are attracted to the injured neutrophils by an enzyme released by the latter (step 32). Phagocytosis of neutrophils by monocytes (step 33) was observed in our studies of C-induced pleurisy (20). This phagocytosis is thought to account for the absence of neutrophils in sections taken 2 days after the interdermal injection of C (8). The interaction of monocytes with neutrophils is associated with the cessation of edema formation (step 34) and the subsequent passive decay of this fluid (20) (step 35). Decay of dermal hyperemia (step 36) is also associated with the mobilization of monocytes and an unknown action of these cells, probably phagocytosis of neutrophils. It is speculated that a factor released from injured endothelium is responsible for the intense dermal fibroblast and vascular proliferation observed histologically in the 2- to 8-day period after injection of C (step 37). The only sign of an inflammatory reaction in the dermis after day 8 is the presence of monocytes. These cells may be incorporated in the new tissue as macrophages (step 38). The dermis appears normal on day 15 (step 39).

Histological changes were also observed in the epidermis after the subplantar injection of C (8, 9). These observations by Militzer have not been studied mechanistically. Therefore, the pathway scheme representing the epidermal response to C (Fig. 9, steps

- Step 29. Release of agent from injured neutrophils which changes configuration of dermal endothelium, t_7-t_{12} h.
- Step 30. Monocytes adhere to dermal endothelium, t_7-t_{12} h.
- Step 31. Diapedesis of adhered monocytes into dermis, t_7-t_{12} h.
- Step 32. Chemotactic cytoplasmic enzyme released from injured neutrophils attracts monocytes, t_7-t_{12} h.
- Step 33. Phagocytosis of neutrophils by monocytes in dermis, t_7-t_{48} h.
- Step 34. Cessation of PI edema formation in dermis, t_7-t_{24} h.
- Step 35. Passive decay of PI edema, t_7-t_{24} h.
- Step 36. Decay of dermal hyperemia, $t_{24}-t_{72}$ h.
- Step 37. Stimulation of fibroblast and vascular tissue proliferation in dermis by factor released from injured endothelium, t_2-t_8 d.
- Step 38. Monocytes incorporated into new dermal tissue, t_8-t_{15} d.
- Step 39. Dermal tissue back to normal, t_{15} d.

Figure 8. Pathway scheme representing the monocytic phase of the dermal PIR to hind-limb injection of C.

40-43) is limited to a listing, in chronological order, of the histological changes he observed. There was no mention of the presence of mobilized phagocytes in the epidermal response to C. Thus, the edema and other changes have been categorized as an NPIR. The epidermal tissue returned to normal 20 days after the injection of C.

Pathway studies of the inflammatory response to subplantar injection of C were undertaken with the expectation of finding this model to be simpler than the C pleurisy model, which consisted of a dual PIR (20). This was not the case. Examination of Figs. 6-9 reveals the hind-limb response to consist of a dual NPIR (dermis and epidermis) and a PIR in the dermis.

BRIEF APPRAISAL OF PATHWAY SCHEME

The sequential 43-step pathway scheme just presented contains several improvements over previous schemes (17, 20). These improvements include the epidermal and algesic changes produced by C. Another improvement is the separation of the overall response into components belonging to either an NPIR or a PIR. This separation

has increased our understanding of the sequence of changes observed microscopically. A conceptual change concerns the mechanism responsible for the localization of phagocytes at the site of irritation. IL 1-induced endothelial cell adhesivity has replaced the concept of the presence of a chemotactic factor in the circulating blood to account for the localization of phagocytes. This new concept is consonant with our previous findings.

Quantitative temporal measurements of edema formation and phagocyte mobilization in combination with specific pharmacological intervention have made it possible to obtain a more precise understanding of the actions and events that occur in C-induced inflammation in the rat than is possible in most models of inflamma-

Figure 9. Epidermal NPIR to interdermal injection of C.

- Step 40. Epidermal intercellular edema, t_4-t_{48} h.
- Step 41. Epidermal hyperplasia, $t_{12}-t_{24}$ h.
- Step 42. Epidermal hyperkeratosis, t_7-t_{20} d.
- Step 43. Epidermal tissue back to normal, t_{20} d.

tion. Nevertheless, perusal of the pathway scheme just presented reveals the naiveté of our understanding of

many aspects of this response (i.e., epidermal NPIR, Fig. 9) and the paucity of knowledge concerning the media-

tors responsible for almost all of the actions and events that have been observed. P

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